

Math Camp

Lecture 5

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Why metric spaces?

- A *space* is a set with some additional structure
 - Vector space is a set with a linear structure (addition and scalar multiplication)
 - Probability space is a set with a collection of events and a probability measure
 - Metric space is a set with a notion of distance, a *metric*

Metric spaces

- Let X be a non-empty set and let $d : X \times X \rightarrow \mathbb{R}_+$ satisfy,
 $\forall x, y, z \in X$,
 - $d(x, y) = 0 \Leftrightarrow x = y$
 - *Symmetry*, $d(x, y) = d(y, x)$
 - *Triangle inequality*, $d(x, z) \leq d(x, y) + d(y, z)$
- The pair (X, d) is a *metric space*, with d a *metric* on X .

Continuity and sequential approximation

- Continuous functions preserve nearness: x near $y \Rightarrow f(x)$ near $f(y)$
 - f is continuous at x if $f(y)$ can be made as close as you like to $f(x)$ by choosing y sufficiently close to x
- Sequential approximation
 - We want to show some property holds for $x \in X$
 - Show it holds for each element of a sequence $\{x_n\}$
 - Show $x_n \rightarrow x$ and that the property respects limits

Normed linear spaces

- A *normed linear space* is a vector space coupled with a *norm* that measures the lengths of its vectors
- If $\mathbf{V}$ is a vector space then a *norm* on $\mathbf{V}$ is a map $\|\cdot\| : \mathbf{V} \rightarrow \mathbb{R}_+$ such that
 - *Positivity*, $\|\mathbf{v}\| = 0 \Leftrightarrow \mathbf{v} = \mathbf{0}$
 - *Homogeneity*, $\|\alpha\mathbf{v}\| = |\alpha|\|\mathbf{v}\|$, $\alpha \in \mathbb{R}$ or $\mathbb{C}$
 - *Triangle inequality*, $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$
- If $(\mathbf{V}, \|\cdot\|)$ is a normed vector space then $d(\mathbf{v}, \mathbf{w}) = \|\mathbf{v} - \mathbf{w}\|$ is a metric on $\mathbf{V}$.

Normed linear spaces

- The norm of $x \in \mathbb{R}$ is its absolute value $|x|$
 - The absolute value is the distance from the origin
- The *taxicab* norm is defined on

$$(\mathbb{R}^n, \|\cdot\|_1) \text{ where } \|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i|$$

- The *Euclidean* norm is defined on

$$(\mathbb{R}^n, \|\cdot\|_2) \text{ where } \|\mathbf{v}\|_2 = \left(\sum_{i=1}^n v_i^2 \right)^{1/2}$$

- The *maximum* norm is defined on

$$(\mathbb{R}^n, \|\cdot\|_\infty) \text{ where } \|\mathbf{v}\|_\infty = \max_{i=1, \dots, n} |v_i|$$

Normed linear spaces

- The *sup* norm is defined on

$$(C(X), \|\cdot\|_\infty) \text{ where } \|f\|_\infty = \sup_{x \in X} |f(x)|$$

- $C(X)$ is the set of continuous functions from a compact metric space X to $\mathbb{R}$

- The L^2 norm is defined on

$$(L^2(\Omega), \|\cdot\|_2) \text{ where } \|x\|_2 = (\mathbb{E}x^2)^{1/2}$$

- $L^2(\Omega)$ is the set of random variables on a probability space (Ω, μ) with $\mathbb{E}x^2 < \infty$
- The Euclidean norm with the sum replaced by an expectation

Sequences

- A *sequence* in (X, d) is an infinite list of elements of X , often denoted $\{x_n\}$
- $\{x_n\}$ *converges* to x , written $x_n \rightarrow x$, provided $d(x_n, x) \rightarrow 0$ in $\mathbb{R}$
 - $\forall \varepsilon > 0, \exists N > 0$ such that $n \geq N \Rightarrow d(x_n, x) < \varepsilon$
- A convergent sequence tends to a unique point called its *limit*

Sequences

$$x_n = \left(\frac{1}{n}, \frac{1}{n}\right)$$

Bounded sequences

- A sequence $\{\mathbf{x}_n\}$ in $\mathbb{R}^m$ is *bounded* if $\exists M \in \mathbb{R}$ such that $\|\mathbf{x}_n\| \leq M$ for all n
- Every convergent sequence in $\mathbb{R}^m$ is bounded
 - Suppose $\mathbf{x}_n \rightarrow \mathbf{x}$. Take ε : $\exists N$ such that $\|\mathbf{x}_n - \mathbf{x}\| < \varepsilon$ for all $n \geq N$
 - For $n \geq N$, by the triangle inequality

$$\|\mathbf{x}_n\| \leq \|\mathbf{x}_n - \mathbf{x}\| + \|\mathbf{x}\| < \varepsilon + \|\mathbf{x}\|$$

- $M = \max\{\|\mathbf{x}_1\|, \dots, \|\mathbf{x}_{N-1}\|, \varepsilon + \|\mathbf{x}\|\}$

Completeness

- Let $\{x_n\}$ be a sequence in (X, d) .
- $\{x_n\}$ is *Cauchy* provided $\forall \varepsilon > 0, \exists N > 0$ such that $m, n > N \Rightarrow d(x_n, x_m) < \varepsilon$
 - The terms get, and stay, close together
- (X, d) is *complete* provided Cauchy sequences converge
 - If $\{x_n\}$ is Cauchy then $\exists x \in X$ such that $x_n \rightarrow x$

Intermediate value theorem

Intermediate Value Theorem

- If $f : [a, b] \rightarrow \mathbb{R}$ is continuous and $f(a) < 0 < f(b)$ (or vice-versa) then $\exists x \in [a, b]$ such that $f(x) = 0$.

Fixed Point Theorem

- A *fixed point* of $f : [a, b] \rightarrow [a, b]$ is $x \in [a, b]$ such that $f(x) = x$
- By the IVT, if f is continuous then it has a fixed point
 - Suppose $f(a) > a$ and $f(b) < b$ (otherwise f has a fixed point) and let $g(x) = f(x) - x$. Then g is continuous with $g(a) > 0$, $g(b) < 0$
- Extensions of the IVT to more general metric spaces come in the form of fixed point theorems

Intermediate value theorem

limsup and liminf

- Let $\{x_n\}$ be a sequence of real numbers.
 - $\limsup x_n = \lim_{n \rightarrow \infty} \sup\{x_k : k \geq n\}$
 - $\liminf x_n = \lim_{n \rightarrow \infty} \inf\{x_k : k \geq n\}$
- Let $\bar{x}_n = \sup\{x_k : k \geq n\}$ and $\underline{x}_n = \inf\{x_k : k \geq n\}$.
- $\{\bar{x}_n\}$ is decreasing and $\{\underline{x}_n\}$ is increasing
- If $\{x_n\}$ is bounded, i.e., there is some interval $[-M, M] \subseteq \mathbb{R}$ such that $x_n \in [-M, M]$ for all n
 - $\{\bar{x}_n\}$ bounded below and $\{\underline{x}_n\}$ bounded above
 - By completeness of $\mathbb{R}$, both sequences converge, i.e. $\limsup x_n$ and $\liminf x_n$ exist

limsup and liminf

$$x_k = \{(1, 1), (1, -1), (-1, 1), (-1, -1), (1, 1), (1, -1), (-1, 1), \dots\}$$

Subsequences

- $\limsup$ and $\liminf$ concern subsequences.
- Let $k : \mathbb{N} \rightarrow \mathbb{N}$ be strictly increasing. The *subsequence* $\{x_{k(n)}\}$ of $\{x_n\}$ has n th element $x_{k(n)}$
 - $\{x_2, x_4, x_6, \dots\}$ is the subsequence with $k(n) = 2n$
- A given $\{x_n\}$ need not converge
 - $y_n = n$ diverges to ∞
 - $x_n = (-1)^n$ bounces between ± 1 forever
- Bounded sequences in $\mathbb{R}$ have convergent subsequences
 - $\liminf$ and $\limsup$ identify the smallest and largest limits of these subsequences

Subsequences

- If $\{x_n\}$ is a bounded sequence of real numbers then
 - $\limsup x_n$ and $\liminf x_n$ exist and $\liminf x_n \leq \limsup x_n$
 - There is a subsequence of $\{x_n\}$ converging to $\limsup x_n$
 - There is a subsequence of $\{x_n\}$ converging to $\liminf x_n$
 - If $x \in \mathbb{R}$ is the limit of a subsequence of $\{x_n\}$ then $\liminf x_n \leq x \leq \limsup x_n$

Subsequences

$$x_n = (-1)^n \left(1 + \frac{1}{n}\right) : \{-2, 1.5, -1.33, 1.25, -1.2, 1.17, \dots\}$$

Bolzano–Weierstrass

Bolzano–Weierstrass Theorem

Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

- In $\mathbb{R}$, take the subsequence converging to $\limsup x_n$
- In $\mathbb{R}^n$, work one coordinate at a time
 - Extract a subsequence along which the first coordinate converges, then a further subsequence along which the second converges, and so on
 - After n steps every coordinate converges, so the subsequence converges

Open and closed sets

- The *ball* of radius ε around $x \in X$ is

$$B(x, \varepsilon) = \{y \in X \mid d(x, y) < \varepsilon\}$$

- $S \subseteq X$ is *open* provided $\forall x \in S$ there is some ball around x contained in S
- $S \subseteq X$ is *closed* provided its complement $S^C = X \setminus S$ is open
- Open and closed sets generalize open and closed intervals

Adherence and closure

- Let $S \subseteq X$. A point $x \in X$ is *adherent* to S if there is a sequence of points of S tending to x
 - Equivalently, every ball $B(x, \varepsilon)$ contains a point of S
 - Any $x \in S$ is adherent to S
- The set of points of X adherent to S is the *closure (adherence)* of S , denoted $\overline{S}$
 - $S \subseteq \overline{S}$
 - For any $A \subseteq X$, $S \subseteq A$ implies $\overline{S} \subseteq \overline{A}$
- S is closed iff $\overline{S} = S$
 - A set is closed iff it contains the limit of every convergent sequence of its points

Interior and boundary

- $x \in X$ is *interior* to S if it is not adherent to S^C
 - Equivalently, some ball around x is contained in S
 - S is open iff every point of S is interior to S
- $x \in X$ is a *boundary point* of S if it is adherent to both S and S^C
 - The *boundary* of S is the set of its boundary points
 - S is closed iff it contains its boundary

Compactness

- $S \subseteq X$ is *bounded* if $S \subseteq B(x, r)$ for some $x \in X$ and $r > 0$
- $S \subseteq X$ is *compact* provided any sequence in S has a subsequence that converges to some point in S
- Intuitively, a compact set
 - is not too big: otherwise it could have sequences whose points stay far from each other
 - contains its boundary: otherwise a sequence could converge to a boundary point outside S , and so would every subsequence
- In any metric space, a compact set is closed and bounded

Compactness

- In $\mathbb{R}^n$ the converse holds

Heine–Borel Theorem

The set $S \subseteq \mathbb{R}^n$ is compact if and only if it is closed and bounded.

- If S is bounded then every sequence in S has a convergent subsequence (Bolzano–Weierstrass)
- If S is closed then the limit of that subsequence stays in S

Compactness

- Compact sets constrain the behavior of continuous functions
- If $f : S \rightarrow \mathbb{R}$ is continuous on a compact set $S \subseteq X$
 - f realizes its maximum and minimum on S so that

$$a_{\max} = \arg \max_{x \in S} f(x) \quad \text{and} \quad a_{\min} = \arg \min_{x \in S} f(x)$$

- If we restrict attention to compact choice sets then agents' optimization problems have solutions